Osama Ghufran

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Summary

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GATE 2025 qualified (AIR 2892) M.Tech Software Engineering student passionate about LangChain, LLMs, and applied Machine Learning. Experienced in building end-to-end AI systems spanning RAG, NLP, deep learning, and REST APIs, with strong problem-solving and debugging skills.

Work Experience

Teaching Assistant, DTU Aug 2025 - present

- Assisted faculty with lectures and doubt-resolution in core CS subjects, and reviewed student code submissions for correctness, edge cases, and debugging quality.
- Guided students through systematic root-cause debugging, strengthening their programming and problem-solving fundamentals.

Projects

AirAssist AI – LangGraph-Based Agentic RAG Assistant GitHub

- Built an AI-powered airline assistant using LangGraph, LangChain, FastAPI, Ollama, and ChromaDB, combining document-based question answering with real-time flight, weather, and currency tools.
- Implemented a hybrid RAG pipeline with semantic vector search, BM25, multi-query expansion, Cross-Encoder reranking, and context compression to improve retrieval relevance.
- Designed stateful LangGraph workflows with conditional routing and conversational memory for context-aware multi-turn airline interactions.
- Developed REST APIs and an interactive web interface with tool error handling, fallback workflows, and execution tracking for reliable airline-related queries.

YouTube Sentiment Analyzer GitHub

- Built an end-to-end NLP pipeline using the YouTube Data API v3 and VADER sentiment analysis to classify up to 100 comments per video as positive, neutral, or negative.
- Integrated a locally hosted SmolLM2-1.7B-Instruct model to auto-generate sentiment verdicts and key discussion themes, deployed via an interactive Gradio dashboard with Matplotlib/Plotly visualizations.

Sign Language Recognition Using Machine Learning GitHub

- Developed a CNN to classify American Sign Language (ASL) alphabet gestures on 27K+ labeled images from the ASL MNIST dataset, achieving 97% validation accuracy using TensorFlow and Keras.
- Performed image preprocessing, model training, and evaluation with confusion matrices to analyze misclassifications and assess class-level recognition performance.

Education

2025 - present	M.Tech Software Engineering at **Delhi Technological University**
2019 - 2023	B.Tech Computer Science at **Jamia Hamdard University**
2018	Class 12th, CBSE Board at **St. Anthony Sr. Sec. School**
2016	Class 10th, CBSE Board at **St. Anthony Sr. Sec. School**

Skills

Programming	Python, NumPy, Pandas
Machine Learning / Deep Learning	TensorFlow, Keras, CNNs, Model Evaluation & Confusion Matrices
Generative AI / NLP	LangChain, LangGraph, RAG, Ollama, HuggingFace Transformers, VADER, BM25, Cross-Encoder Reranking, ChromaDB
Backend & Deployment	FastAPI, REST APIs, Gradio
Visualization	Matplotlib, Plotly